Machine Learning Adversarial Attacks

Internship Evaluation

Understanding Security Vulnerabilities in AI Systems

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CSE AI 2

Contents

What Are Adversarial Attacks?

Core Concept

Adversarial attacks are deliberate perturbations added to input data that cause machine learning models to make incorrect predictions while appearing unchanged to humans.

Stealth Nature

Imperceptible changes that fool AI systems

Security Risk

Critical vulnerabilities in deployed models

Attack Classification Matrix

White-box Attacks

Full model knowledge: gradients, architecture, parameters

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Black-box Attacks

Limited knowledge: query-based, transfer attacks

Non-targeted Attacks

Any incorrect prediction suffices

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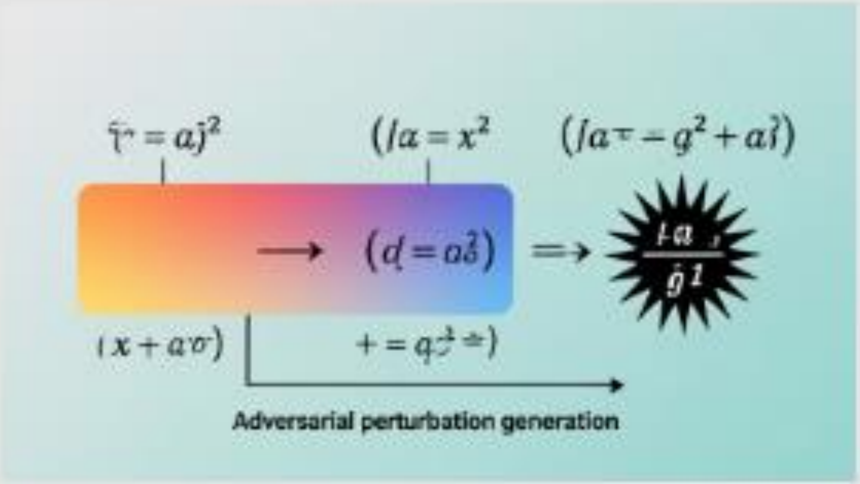
Targeted Attacks

Specific misclassification goals

Fast Gradient Sign Method (FGSM)



Gradient Computation

Calculate loss gradient with respect to input data

Perturbation Generation

Apply sign of gradient with small epsilon

Adversarial Creation

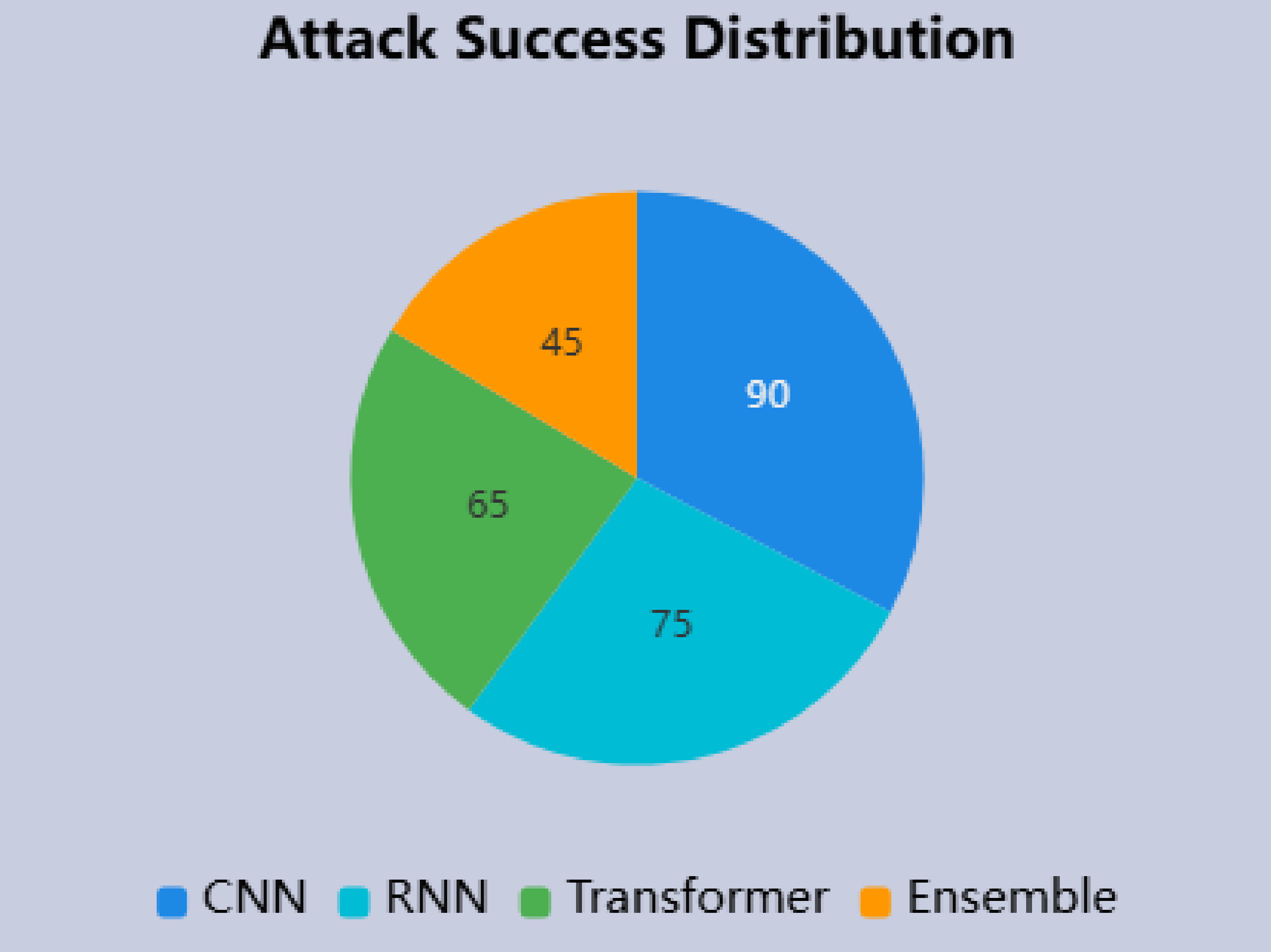
Add perturbation to original input

Attack Success Rates by Model Type

Vulnerability Analysis

Deep neural networks show varying susceptibility to adversarial attacks across different architectures and training methods.

> CNNs: 85-95% attack success rate, RNNs: 70-80%, Transformers: 60-75%



Attack Generation Pipeline

Input Selection

Choose clean sample from target dataset

Gradient Analysis

Compute model's sensitivity to input perturbations

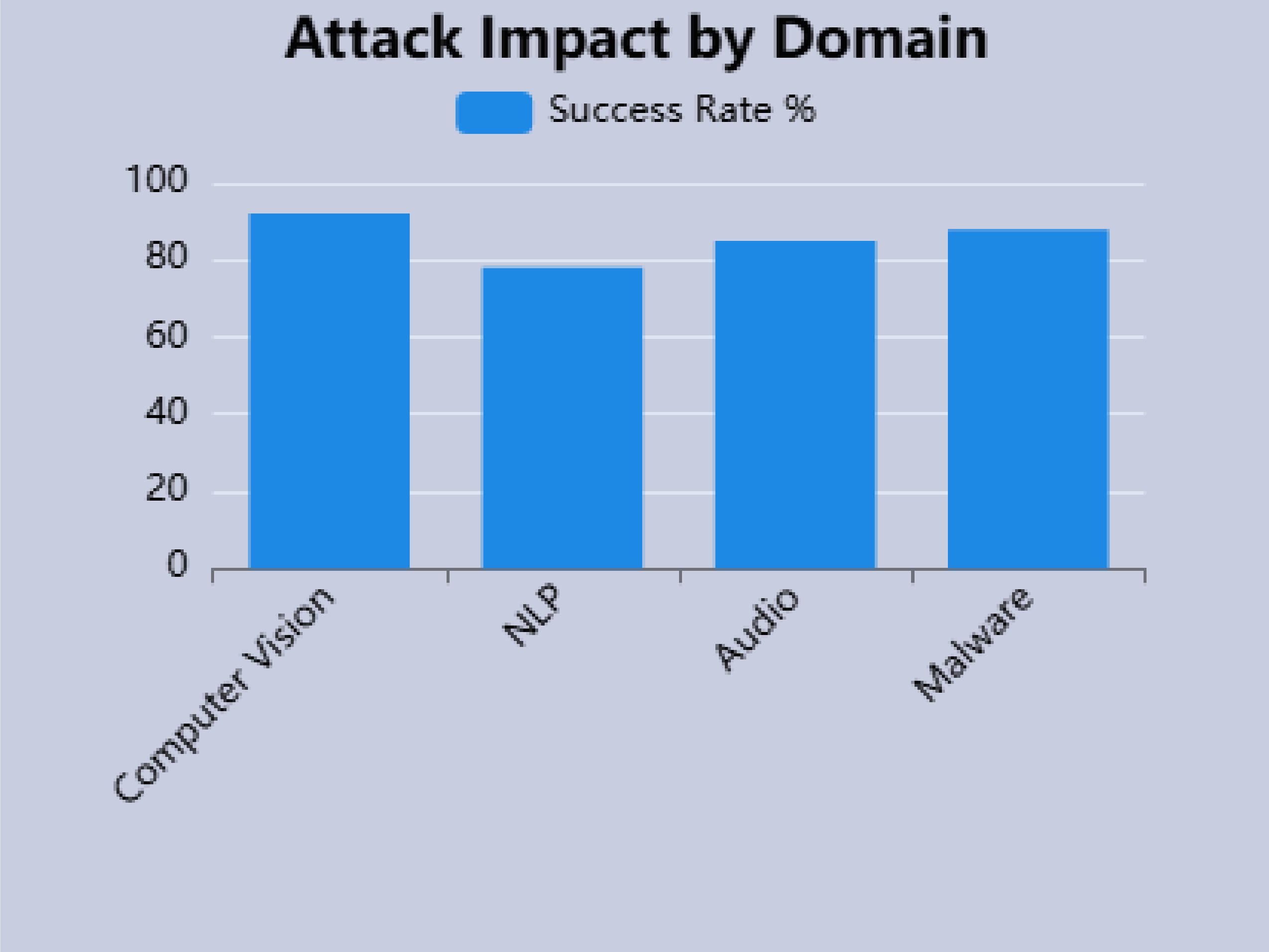
Perturbation Crafting

Generate minimal adversarial noise using optimization

Validation Testing

Verify attack effectiveness and imperceptibility

Real-World Attack Scenarios



Autonomous Vehicles

Stop sign misclassification causing safety risks

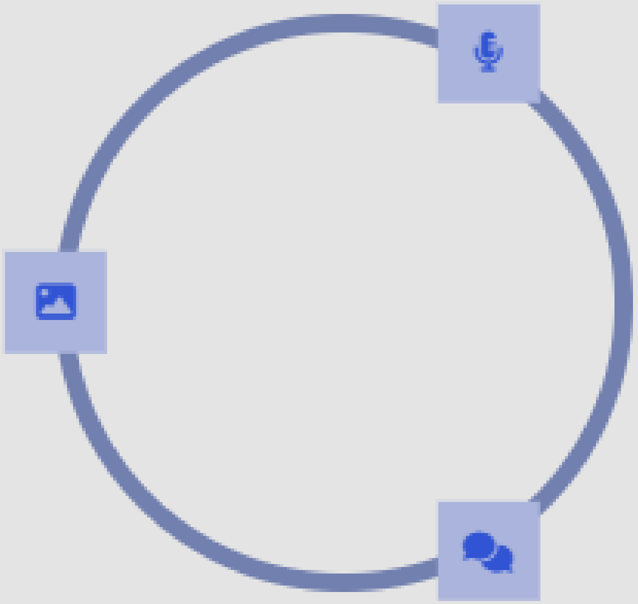
Medical Imaging

Cancer diagnosis manipulation with subtle pixel changes

Defense Strategy Framework

Adversarial Training

Incorporate adversarial examples during model training to improve robustness against known attack patterns



Input Transformations

Apply preprocessing techniques like denoising, compression, or randomization to reduce perturbation effects

Certified Defenses

Mathematical guarantees of robustness within specified perturbation bounds using formal verification methods

Defense Effectiveness Metrics

85%
Adversarial Training

72%
Input Denoising

94%
Ensemble Methods

91%
Certified Robustness

Adversarial Training

Most practical defense method

- Retrains on adversarial examples
- Improves robustness significantly
- Computationally expensive

Certified Defenses

Strongest theoretical guarantees

- Provides mathematical guarantees
- Limited to small perturbations
- High computational complexity

Detection Methods

Complementary to robust training

- Identifies adversarial inputs
- Rejects suspicious samples
- May increase false positives

Future Research Directions

Emerging Challenges

Next-generation adversarial attacks and defenses require addressing more sophisticated threat models and real-world deployment constraints

Physical Attacks

Adversarial patches in real-world environments

Universal Perturbations

Single perturbation across multiple inputs

Adaptive Attacks

Attacks that evolve against new defenses

Thank You

Questions and Discussion Welcome